

Year 10 Science (Adjusted)
Semester 1 Examination Revision 2019
Physical Sciences

1. What is the difference between **distance** and **displacement**? Draw a diagram to help your explanation.

DISTANCE: movement made in any direction.

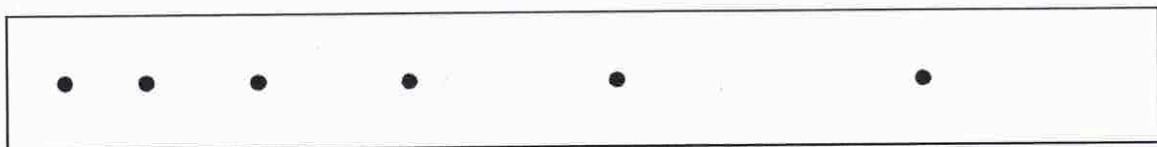
DISPLACEMENT: measurement from the start to the finish in a straight line in a particular direction.

2. Complete the following table. Make sure you include the appropriate units for each quantity.

Distance travelled	Time taken	Average speed
400 m	45 s	<i>8.89 m/s</i>
3000 km	<i>5 hrs</i>	600 km/h
<i>750 m</i>	25 s	30 m/s
200 m	<i>8 s</i>	25 m/s

3. For each ticker tape shown, describe the motion of the body to which the tape was attached. The left-hand dot is at time zero.

(a)



speeds up (accelerates)

(b)



speeds up (accelerates), then slows down (decelerates).

4. A man drives his car 800 m due East and then turns North, driving 300 m before stopping at a petrol station. It took him 120 s to get there.

(a) What distance he has driven.

$$d = 800 + 300$$

$$= \underline{1100 \text{ m}}$$

(b) Calculate his average speed for the journey.

$$sp = ?$$

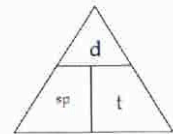
$$d = 1100 \text{ m}$$

$$t = 120 \text{ s}$$

$$sp = \frac{d}{t}$$

$$= \frac{1100}{120}$$

$$= \underline{9.17 \text{ m/s}}$$



5. A balloon filled with hydrogen gas has a velocity of 40 m/s East. How long will it take to go 450 m ?

$$sp = 40 \text{ m/s}$$

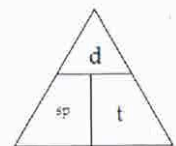
$$d = 450 \text{ m}$$

$$t = ?$$

$$t = \frac{d}{sp}$$

$$= \frac{450}{40}$$

$$= \underline{11.2 \text{ s}}$$



6. What is the average speed of a plane if it flies 2700 km in 2.3 hours? Give the answer in km/h.

$$sp = ?$$

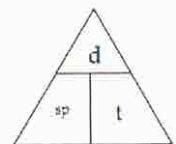
$$d = 2700 \text{ m}$$

$$t = 2.3 \text{ hrs}$$

$$sp = \frac{d}{t}$$

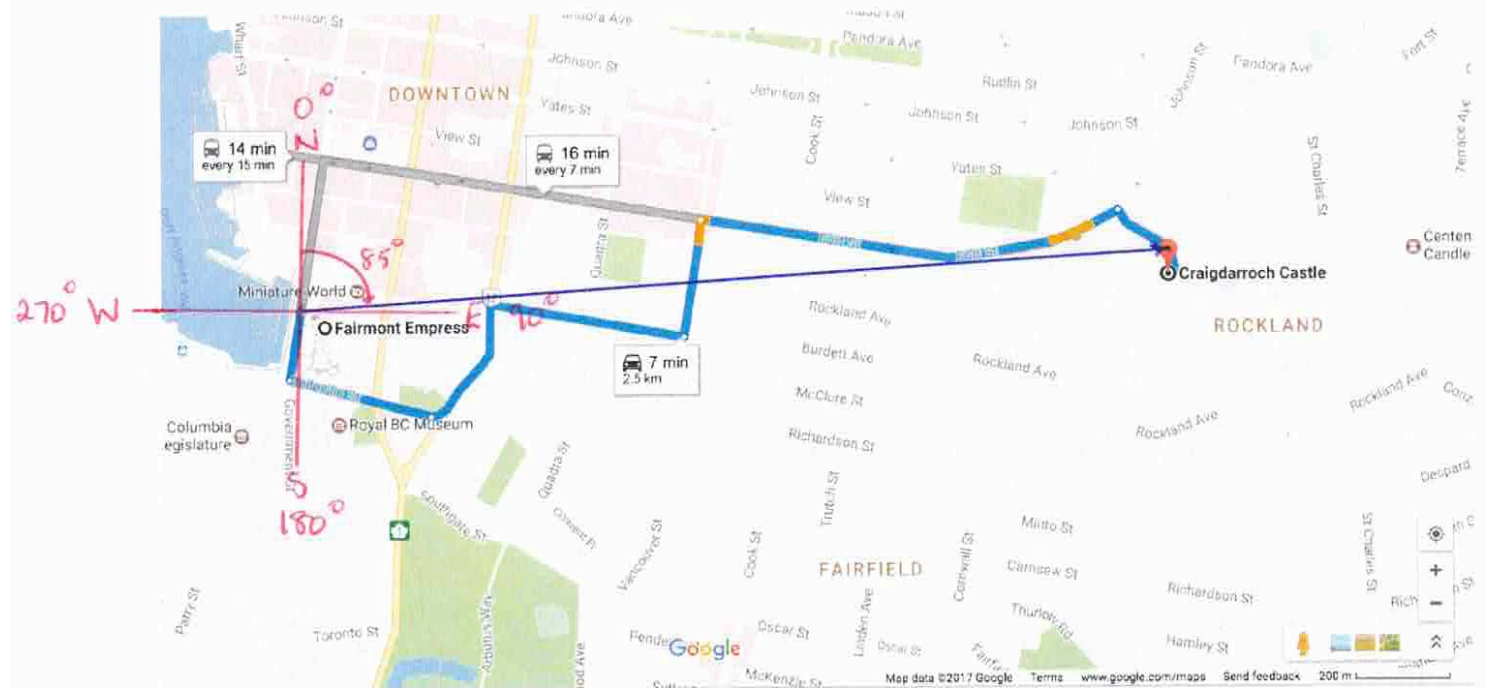
$$= \frac{2700}{2.3}$$

$$= \underline{1174 \text{ km/h}}$$



7. Use the scale in the map below to work out the **distance and displacement** from the Fairmont Empress hotel to Craigdarroch Castle. Use either the blue or grey route.

Remember to give a bearing or direction for the displacement.



distance: 2.5 km (blue path)

displacement: 1900 m bearing 85°
(± 50m) (± 2°)

$$1.2 \text{ cm} = 200 \text{ m}$$

$$\text{displacement} = \frac{11.4}{1.2} \times \frac{200}{1} = 1900 \text{ m}$$